

1 Introduction

In chapter we introduce the notion of a proof and describe methods for constructing proofs. A proof is a valid argument that establishes the truth of a mathematical statement. A proof can use the hypotheses of the theorem, if any, axioms assumed to be true, and previously proven.

The methods of proof discussed in this chapter are important not only because they are used to prove mathematical theorems, but also for their many applications to computer science. These applications include verifying that computer programs are correct, establishing that operating systems are secure, making inferences in artificial intelligence, showing that system specifications are consistent, and so on. Consequently, understanding the techniques used in proofs is essential both in mathematics and in computer science.

2 Terminology

- **Theroem** : is a statement that can be shown to be true. We demonstrate that a theorem is true with a proof. A proof is a valid argument that establishes the truth of a theorem
- **Proposition** : less important theorems
- **Lemma** : A less important theorem that is helpful in the proof of other results. Complicated proofs are usually easier to understand when they are proved using a series of lemmas, where each lemma is proved individually
- **Corollary** : is a theorem that can be established directly from a theorem that has been proved.
- **Conjecture** : is a statement that is being proposed to be a true statement, usually on the basis of some partial evidence, a heuristic argument, or the intuition of an expert.

3 Methods of Proving Theorems

3.1 Direct Proofs

A direct proof of a conditional statement $p \longrightarrow q$ is constructed when the first step is the assumption that p is true; subsequent steps are constructed using rules of inference, with the final step showing that q must also be true. A direct proof shows

that a conditional statement $p \longrightarrow q$ is true by showing that if p is true, then q must also be true, so that the combination p true and q false never occurs. In a direct proof, we assume that p is true and use axioms, definitions, and previously proven theorems, together with rules of inference, to show that q must also be true.

Definition 1. The integer n is even if there exists an integer k such that $n = 2k$, and n is odd if there exists an integer k such that $n = 2k + 1$. (Note that every integer is either even or odd, and no integer is both even and odd.) Two integers have the same parity when both are even or both are odd; they have opposite parity when one is even and the other is odd.

Example 1 : Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."

Note that this theorem states $\forall n P(n) \rightarrow Q(n)$, where $P(n)$ is " n is an odd integer" and $Q(n)$ is " n^2 is odd". To begin a direct proof of this theorem, we assume that the hypothesis of this conditional statement is true, namely, we assume that n is odd. By the definition of an odd integer, it follows that $n = 2k + 1$, where k is some integer.

We want to show that n^2 is also odd. We can square both sides of the equation $n = 2k + 1$ to obtain a new equation that expresses n^2 . When we do this, we find that $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. By the definition of an odd integer, we can conclude that n^2 is an odd integer (it is one more than twice an integer). Consequently, we have proved that if n is an odd integer, then n^2 is an odd integer.

Example 2 : Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square. (An integer a is a perfect square if there is an integer b such that $a = b^2$.)

3.2 Proof by Contraposition

An extremely useful type of indirect proof is known as proof by contraposition. Proofs by contraposition make use of the fact that the conditional statement $p \rightarrow q$ is equivalent to its contrapositive, $\neg q \rightarrow \neg p$. This means that the conditional statement $p \rightarrow q$ can be proved by showing that its contrapositive, $\neg q \rightarrow \neg p$, is true. In a proof by contraposition of $p \rightarrow q$, we take $\neg q$ as a premise, and using axioms, definitions, and previously proven theorems, together with rules of inference, we show that $\neg p$ must follow. In an other word, a proof by contraposition consists of a direct proof of the contrapositive $\neg q \rightarrow \neg p$.

Example 1 : Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Example 2 : Prove that if n is an integer and n^2 is odd then n is odd.

3.3 Proof by Contradiction

Suppose we want to prove that a statement p is true. Furthermore, suppose that we can find a contradiction q such that $\neg p \rightarrow q$ is true. Because q is false, but $\neg p \rightarrow q$ is true, we can conclude that $\neg p$ is false, which means that p is true. How can we find a contradiction q that might help us prove that p is true in this way?

Because the statement $r \wedge \neg r$ is a contradiction whenever r is a proposition, we can prove that p is true if we can show that $\neg p \rightarrow (r \wedge \neg r)$ is true for some proposition r . Proofs of this type are called proofs by contradiction.

Example 1 : Prove that $\sqrt{2}$ is irrational by giving a proof by contradiction.

Example 2 : Give a proof by contradiction of the theorem "If $3n + 2$ is odd, then n is odd."

3.4 Proofs of Equivalence

To prove a theorem that is a biconditional statement, that is, a statement of the form $p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$ are both true. The validity of this approach is based on the tautology : $(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$.

Example 1 : Prove the theorem "If n is an integer, then n is odd if and only if n^2 is odd."

Example 2 : Show that these statements about the integer n are equivalent :

- $p_1 : n$ is even
- $p_2 : n - 1$ is odd
- $p_3 : n^2$ is even

3.5 Proofs by Counter Example

We stated previously that to show that a statement of the form $\forall x P(x)$ is false, we need only find a counterexample, that is, an example x for which $P(x)$ is false.

Example 1 : Show that the statement "Every positive integer is the sum of the squares of two integers" is false.

3.6 Exhaustive Proof and Proof by Cases

Sometimes we cannot prove a theorem using a single argument that holds for all possible cases. We now introduce a method that can be used to prove a theorem, by considering different cases separately. This method is based on a rule of inference that we will now introduce. To prove a conditional statement of the form $(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q$ the tautology $[(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q] \leftrightarrow [(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)]$ can be used as a rule of inference.

3.6.1 Exhaustive Proof

Some theorems can be proved by examining a relatively small number of examples. Such proofs are called exhaustive proofs, or proofs by exhaustion because these proofs proceed by exhausting all possibilities. An exhaustive proof is a special type of proof by cases where each case involves checking a single example. We now provide some illustrations of exhaustive proofs.

Example 1 : Prove that $(n + 1)^3 \geq 3^n$ if n is a positive integer with $n \leq 4$.

3.6.2 Proof by Cases

A proof by cases must cover all possible cases that arise in a theorem. We illustrate proof by cases with a couple of examples. In each example, you should check that all possible cases are covered.

Example 1 : Prove that if n is an integer, then $n^2 \geq n$.

Example 2 : Use a proof by cases to show that $|xy| = |x||y|$, where x and y are real numbers.

3.7 Existence Proofs

Many theorems are assertions that objects of a particular type exist. A theorem of this type is a proposition of the form $\exists x P(x)$, where P is a predicate. A proof of a proposition of the form $\exists x P(x)$ is called an existence proof. There are several ways to prove a theorem of this type. Sometimes an existence proof of $\exists x P(x)$ can be given by finding an element a , called a witness, such that $P(a)$ is true. This type of existence proof is called constructive. It is also possible to give an existence proof that is nonconstructive; that is, we do not find an element a such that $P(a)$ is true, but rather prove that $\exists x P(x)$ is true in some other way.

Example 1 : A Constructive Existence Proof : Show that there is a positive integer that can be written as the sum of cubes of positive integers in two different ways.

Example 2 : A Nonconstructive Existence Proof : Show that there exist irrational numbers x and y such that x^y is rational.

4 Exercises

Exercise 1 :

1. Show that if $n^2 - 3$ is even, then n is odd. Proof by contrapositive.
2. Show that for any real number x , if $x + 2 > 10$, then $x^2 - 14 > 5x$. Use direct proof.
3. Show by contradiction $\forall x \in \mathbb{R} : x^2 - 5x \leq 0 \longrightarrow (x \geq 0) \wedge (x \leq 5)$
4. Prove that for any positive integer n , n is odd if and only if $5n + 6$ is odd.
5. For any real number x , if $-4 \leq x \leq 4$, then $x^2 + x - 20 \leq 0$. Prove by contrapositive.
6. Prove By contradiction that for any integer a : if $a^2 - 1$ is odd , then a is even.
7. Prove by cases, if m is an integer, then $m^2 + m + 1$ is always odd.
8. Prove that if n is a positive integer, then n is even if and only if $7n + 4$ is even.
9. Suppose that $x, y \in \mathbb{R}$.Prove that if $y^3 + yx^2 \leq x^3 + xy^2$ then $y \leq x$.
10. Prove that if $x^2 + 2x - 15 \geq 0$ then $x > 3$ or $x < -5$.
11. Show that if $n^2 - 3$ is even, then n is odd. proof by contrapositive
12. For any real number x , if $x + 2 \geq 10$, then $x^2 - 14 \geq 5x$. Use direct proof. Use cases.
13. Prove that if $n \geq 5$ then $n^2 - 7n + 12 \geq 0$.
14. Proof that the number $2t - 17$ is odd for any integer t
15. Proof that For any real number x , if $x^2 + x + 1 < 13$, then $x < 3$. Hint : What is the contrapositive statement ?
16. Given that $\sqrt{2}$ is irrational. Show that $\sqrt{2} + n$ is irrational for any integer n . Use Proof by Contradiction.
17. Show that (proof by cases) if an integer n is not divisible by 3, then $n^2 = 3k + 1$ for some integer k
18. Given r an irrational number. Proof that $2r$ is irrational.
19. Prove by contrapositive and cases that : for any integer n , if $n^2 - 1$ is not divisible by 3 then n is divisible by 3.
20. For any integer n , show that if n is not divisible by 3 , then $n^2 - 9$ is not divisible by 3.
21. Show that $n(n^2 - 1)(n + 2)$ is divisible by 4 for all integer n .
22. Proof that, for any integers a and b , if $a^2 + b^2$ is divisible by 4, then either a is not odd or b is not odd.
23. Use Direct Proof to show that for any integers a and b , if $2a + b$ is divisible by 3, then $5b + a$ is divisible by 3.

Exercise 2 :

Give a counter example for each :

1. $\forall n \in \mathbb{Z} \frac{n}{n+1}$ is not integer
2. $\forall a, b, c \in \mathbb{Z} a|(b+c) \rightarrow (a|b) \vee (a|c)$

Exercise 3 :

Give an example for each :

1. $\exists a \in \mathbb{Z}^+ \frac{a^2+4}{a}$ is perfect square.
2. $\exists a \in \mathbb{Z} a^2 + 3a + 2 < 0$ and $a < 0$.

Exercise 4 :

State by True or False. Give example or counterexample.

1. $\exists x, y \in \mathbb{Z} : x^2 - xy + 1$ is a perfect square.
2. $\forall n \in \mathbb{Z}^+ : n^2 - n + 3$ is prime.
3. $\forall a, b \in \mathbb{Z} : \sqrt{a+b} < a \longrightarrow (\sqrt{a} + \sqrt{b}) < a$
4. $\forall a, b, c \in \mathbb{Z} : ab|(b+c) \longrightarrow ac|(b+c)$